Yuexin (Rachel) Li

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Education

New York University New York, NY
Master of Arts in General Psychology *Spring 2026 – Fall 2027 (expected)*

Columbia University New York, NY
Master of Science in Applied Analytics *Fall 2021 – Fall 2022*
GPA: 3.8/4.0

- Master's Thesis: "An Unsupervised–Supervised Pipeline for Music Streaming: Birch/DBSCAN Clustering and Track Behavior Prediction"

Syracuse University Syracuse, NY
Bachelor of Arts in Psychology, Bachelor of Science in Accounting *Fall 2017 – Spring 2021*
GPA: 3.42/4.0; Honors: Cum Laude

- Senior Thesis: "Cognitive Aging and Financial Decision-Making: A Review"

Research Interests

Visual Perception and Cognition; Mental Representation; Computer Vision; Machine Learning; Attention; Decision-Making; Psycholinguistics

Awards

Cognitive Development Society Diversity Award Feb 2026
Graduate Student Conference Presentation Award NYU Psychology, Feb 2026

Research Experience

Research Assistant Feb 2026 – Present
Landy Lab for Vision and Action
Principal Investigator: Prof. Michael S. Landy New York University

Project 1: Confidence Judgments in Audiovisual Spatial Integration - Supervisor: Luhe Li

- Collect behavioral data and run audiovisual spatial integration experiments.

Research Assistant May 2023 – Dec 2025
Language and Cognitive Neuroscience Lab
Principal Investigator: Prof. Peter Gordon Teachers College, Columbia University

Project 1: Eye-Tracking Study on Infant Event Representation

- Used eye-tracking to investigate whether pre-linguistic infants can form event representations for simple social interaction such as Giving, Hugging, and Showing.
- Developed *Eye-Track-ML*, a machine-learning pipeline (using YOLOv11 for object detection, SAM 2.1 for object segmentation) to automate frame-by-frame annotation for eye-tracking videos and completed annotations for large-scale datasets (>600,000 frames).
- Conducted gaze pattern analyses and visualizations (e.g., heatmaps) to quantify looking time across Areas of Interest (AOIs).

Project 2: Electroencephalogram (EEG) Numbers Study - Supervisor: Dr. Jean Tang

- Used 128-channel EEG to examine the neurobehavioral basis of differentiation between small (1–3) vs. large (4–6) numerosity and change directionality (increase/decrease).
- Adapted the Harvard Automated Processing Pipeline for Electroencephalography (HAPPE) method; performed cleaning, filtering, and segmentation on EEG data and sorted data based on stimulus conditions.
- Preprocessed EEG and Behavioral datasets for analysis and conducted statistical analyses using general linear models, correlation, repeated measures ANOVA.

- Conducted Event-Related Potentials (ERP) analysis by generating ERP signal montages for designated brain areas (e.g., N1 over POT).
- Trained two separate 3-class Convolutional Neural Networks to classify landing digits in the 1–3 vs. 4–6 range; tested whether non-symbolic numbers produce distinct, decodable brain activity patterns.

Project 3: Development of Numerical Cognition in Young Children

- Investigated early-stage development of numerical cognition with a series of tasks that examine acquisition of the meaning of number words in children from different linguistic and cultural backgrounds (English, Mandarin, Spanish).
- Coded experiment videos by capturing children's audio and behavioral responses to different tasks.
- Performed statistical analyses (t-tests, correlation, general linear models) on data from 84 Mandarin-speaking children and generated result visualizations; compared outcomes with a previous study on English-speaking children.

Project 4: Mental State Verb Learning and Theory of Mind in Mandarin-Speaking Children

- Investigated whether Mandarin-speaking children can learn mental state verbs like "think" and "want" through grammatical structure, a process called syntactic bootstrapping.
- Developed analytical methods and analyzed data from 30 Mandarin-speaking children using t-tests and one-way ANOVA; created visualization plots.
- Assisted with experiment video coding and data compilation.

Independent Research

Independent Project Jan 2026 – Apr 2026

Good-Enough Vision at Birth: Newborn Face Information Under Low-Acuity Input

- Developed a computational vision project asking whether newborn-like, rod-dominated, low-acuity visual input contains sufficient information to support early face-related discrimination.
- Built a reproducible Python pipeline using the Chicago Face Database and programmatically generated schematic face stimuli; simulated neonatal input through grayscale conversion, downsampling, Gaussian blur, and edge extraction.
- Evaluated logistic regression, linear SVM, and shallow CNN models on schematic face, face-vs.-phase-scrambled, and attractiveness tasks; found that meaningful face-related information survives severe visual degradation, with attractiveness-related signal showing strong orientation sensitivity.

Conference Presentations

Li, Y. (Apr 2026). **Good-Enough Vision at Birth: A Computational Demonstration of What Newborns Can See with Rod-Dominated, Low-Acuity Input**. Poster accepted for presentation at the 2026 Cognitive Development Society Conference.

Gushiken, M., **Li, Y.**, Gordon, P. (Apr 2026). **How Infant Looking Patterns to Trivalent Events Change from Object to Person Interaction**. Poster accepted for presentation at the 2026 Cognitive Development Society Conference.

Gushiken, M., **Li, Y.**, Tang, J. E., Gordon, P. (Mar/Apr 2025). **Eye-Track-ML: Automated Frame-by-Frame Coding of Eye-Tracking Videos**. Poster presented at the 2025 Columbia AI Summit and 2025 Columbia Data Science Day.

Tang, J. E., **Li, Y.**, Smith, P., Olivares, A. G., Mantaro, E. R., Kirby, E., Bisbee, N., Gordon, P. (Oct 2024; Mar 2025). **Effects of Change Direction and Size on ERP Signatures of Small vs. Large Numerical Perception**. Poster presented at the 2024 APS Global Psychological Science Summit and 2025 Cognitive Neuroscience Society Annual Meeting.

Li, Y., Tang, J. E., Kang, Y., Lin, N., Pei, P., Ye, H., Jin, L., Wang, E., Chang, Y., Liu, T., Zhou, X., Gordon, P. (Oct 2024). **Numerical Cognition in Mandarin-Speaking 2-to-4-Year-Olds in China.** Poster presented at the 2024 APS Global Psychological Science Summit.

Chang, Y., **Li, Y.**, Gao, S., Liu, T., Liu, H., Gordon, P. (May 2024). **Mental State Verb Learning and Theory of Mind in Mandarin-Speaking Children.** Poster presented at the 2024 Association for Psychological Science Annual Convention.

Teaching Experience

Teaching Assistant Spring 2026
New York University: PSYCH-UA8 Data Literacy for Psychology *New York, NY*

- Supported the course instructor by holding office hours, grading assignments/exams, proctoring assessments, and assisting with course administration and student support.

Data Analysis Workshop Instructor Summer 2025
Language and Cognitive Neuroscience Lab *New York, NY*

- Designed and led a series of workshops focused on foundational R programming, statistical methods, and data visualization for research in cognitive neuroscience.

Guest Lecturer Mar 2025
Teachers College BBSN 5022: Eye-Tracking and Dynamic Data Analysis *New York, NY*

- Presented the Eye-Track-ML pipeline; demonstrated automated eye-tracking annotation methods.

R Tutor Fall 2024 – Fall 2025
Language and Cognitive Neuroscience Lab *New York, NY*

- Conducted weekly R training sessions to equip research assistants with skills for data visualization and statistical analysis.

Accounting Tutor Fall 2020
Beta Alpha Psi (Honor Society for Accounting) *Syracuse, NY*

- Held weekly tutoring sessions for students in introductory accounting courses; coordinated with accounting professors to ensure that tutoring materials were current and relevant.

Professional Experience

Data Scientist Intern Feb 2024 – Jun 2024
VisionX LLC *New York, NY*

- Assisted in improving and training advanced stock trading AI models for the data science application team.
- Enhanced data collection procedures by ensuring the acquisition of relevant information for analytics systems; processed, cleaned, and verified data integrity.

Analytical Assistant / Website Designer May 2023 – Feb 2024
Museum of Reclaimed Urban Space *New York, NY*

- Researched and quantified the positive environmental impact of organizations in New York City (e.g., number of community gardens preserved and their local impact).
- Edited and updated the organization's website by refreshing event details, updating content, and embedding multimedia elements.

Consulting Intern Summer 2019
PricewaterhouseCoopers (PwC) China *Shanghai, China*

- Conducted tests and recorded the results of new programs developed by AI teams; documented errors, evaluated performance, and wrote progress reports.

Leadership Experience

Peer Mentor Fall 2018 – Fall 2019
Syracuse University *Syracuse, NY*

- Mentored first-year students, guided mentees through Orientation Week and provided semester-long check-ins to support academic, social, and well-being adjustment.
- Connected mentees to campus supports including Career Center, Wellness Services, Writing Center, and Academic Advising, clarifying policies, deadlines, and services access.

Skills

Programming: R, Python, PostgreSQL, MATLAB
Tools: LaTeX, WordPress, GitHub, Microsoft Office